

Large Language Models for the
Economic and Social Sciences:

AI Safety and Alignment

Indira Sen, HWS 2025

<https://github.com/dess-mannheim/LLM4ESS-HWS25>

Summary of Last Week and Today

Last time we learned:

- Impact of LLM
- Potential Harms of LLMs
- Three LLM-based case studies in the Social Sciences
- Ethical frameworks, principles, and questions

Today:

- Revisiting AI Alignment
- Limits of AI Alignment
- Social Engineering with LLMs
- AI safety issues in social and economic science applications

Recap: Ethical Impacts of AI

First: Which aspects of our live have LLMs impacted?

Long term vs. short term

- Individual
 - Studying, education
 - Communication
 - coding / programming
 - Search
 - Information
 - Job search
 - Coaching and advice
 - Processing
 - Thinking, e.g., brainstorming, bouncing off ideas,
- Decision-making
- Content creation
- Societal impacts
 - Job market
- General laziness vs. efficiency
- More fragmentation, polarization
- Restructuring the internet
 - + widely available information
 - - enshittification
 - - synthetic data degradation
 - - poisoning
- Health diagnoses
- Accelerating science, including biotech
- environmental

Dual Use

- Dual use items: “refer to goods, software and technology that can be used for both civilian and military applications.”
 - More generally, harmful **reuse** of technology, science, and research
- Dual use dilemma: Ammonia, which was useful for agriculture, used for chemical warfare
- Are there dual uses for LLMs?

https://en.wikipedia.org/wiki/Dual-use_technology

Kaffee, Lucie-Aimée, et al. "[Thorny roses: Investigating the dual use dilemma in natural language processing.](#)"

Thorny Roses: Investigating the Dual Use Dilemma in Natural Language Processing

Lucie-Aimée Kaffee¹, Arnav Arora², Zeerak Talat³, Isabelle Augenstein²

¹Hasso Plattner Institute, Germany, ²University of Copenhagen, Denmark

³Mohamed Bin Zayed University of Artificial Intelligence, United Arab Emirates

lucie-aimee.kaffee@hpi.de, aar@di.ku.dk, z@zeerak.org, augenstein@di.ku.dk

Abstract

Dual use, the intentional, harmful reuse of technology and scientific artefacts, is an ill-defined problem within the context of Natural Language Processing (NLP). Large Language Models (LLMs) have advanced capabilities and become more accessible, increasing their potential for intentional misuse beyond their intended use. To prevent such misuse, it is necessary for NLP researchers to understand and document the potential harms of their research. Hence, we propose a specific definition of dual use for NLP researchers and practitioners. We propose a checklist for NLP researchers and practitioners to assess the potential for dual use in their research. The checklist is created based on research and practitioner

research artefacts. This is reflected in contemporary ethical review processes which emphasise the impacts of research on individual subjects, rather than

Area	vulnerability	harms
NLP Applications	4.3	crime, oppression, manipulation
Ethics and NLP	4.1	ethics washing, surveillance, plagiarism, oppression
Psycholinguistics	4.0	cyber bullying, oppression
Generation	3.8	crime, cyber bullying, manipulation, oppression, ethics washing
Dialogue Systems	3.8	surveillance, crime
MT and Multilinguality	3.3	surveillance, crime
ML for NLP	3.2	military application, manipulation, oppression
Resources and Evaluation	3.1	ethics washing, surveillance, manipulation
Interpretability	3.0	ethics washing, manipulation
Information Extraction	2.8	surveillance, censorship
IR and Text Mining	2.7	surveillance

What are the real and potential harms of LLMs?

Table 1 | High-level overview of risks of harm from generative AI systems

Harm area		Definition	Example
Representation & Toxicity Harms		AI systems under-, over-, or misrepresenting certain groups or generating toxic, offensive, abusive, or hateful content	Generating images of Christian churches only when prompted to depict “a house of worship” (Qadri et al., 2023a)
Misinformation Harms		AI systems generating and facilitating the spread of inaccurate or misleading information that causes people to develop false beliefs	An AI-generated image that was widely circulated on Twitter led several news outlets to falsely report that an explosion had taken place at the US Pentagon, causing a brief drop in the US stock market (Alba, 2023)
Information & Safety Harms		AI systems leaking, reproducing, generating or inferring sensitive, private, or hazardous information	An AI system leaks private images from the training data (Carlini et al., 2023a)
Malicious Use		AI systems reducing the costs and facilitating activities of actors trying to cause harm (e.g. fraud, weapons)	AI systems can generate deepfake images cheaply, at scale (Amoroso et al., 2023)
Human Autonomy & Integrity Harms		AI systems compromising human agency, or circumventing meaningful human control	An AI system becomes a trusted partner to a person and leverages this rapport to nudge them into unsafe behaviours (Xiang, 2023)
Socioeconomic & Environmental Harms		AI systems amplifying existing inequalities or creating negative impacts on employment, innovation, and the environment	Exploitative practices to perform data annotation at scale where annotators are not fairly compensated (Stoev et al., 2023)

Ethical Principles to Questions

- **Respect for Persons:** treating people as autonomous and honoring their wishes.
- **Beneficence:** understanding and improving the risk/benefit profile of your study, and then deciding if it strikes the right balance.
- **Justice:** ensuring that the risks and benefits of research are distributed fairly.
- **Respect for Law and Public Interest:** include all relevant stakeholders, not just research participants

Do you treat all research subjects with respect and dignity?

How do you minimize risks and maximize benefits?

Which people benefit from your research? Does it make the powerful more powerful?

What is the impact on people beyond your research subjects, e.g., the general public?

Putting Ethics into Action: Responsible AI and Science

- For research, reproducibility is paramount
- But sometimes, conflict with privacy, especially with sensitive subjects
- Also conflicts with dual use — malicious usage
- Recommendations
 - Share your code
 - Share your data
- If using human subjects, e.g., for interviews or annotations:
 - Pay them fairly (especially if they are not co-authors)
 - Report their demographics (without compromising privacy!) — anonymization

NeurIPS 2021 Paper Checklist Guidelines

The NeurIPS Paper Checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. For each question in the checklist:

- You should answer yes, no, or n/a.
- You should reference the section(s) of the paper that provide support for your answer.
- You can also optionally write a short (1-2 sentence) justification of your answer.

You are encouraged to take a look at the checklist early on as it likely will

<https://neurips.cc/Conferences/2021/PaperInformation/PaperChecklist>

Reviewers will be asked to use the checklist as one of the factors in their decision. It is preferable to answer "no" than "n/a". In general, answering "no" or "n/a" is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgement and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material.

We provide guidance on how to answer each question below. You may additionally refer to our [blog post](#) introducing the checklist to learn more about its motivation and how it was created.

1. For all authors...

- (a) Do the main claim
 - Claims in the paper
 - The paper's conclusion as long as it is clear
- (b) Have you read the paper?
 - Please read the paper

‘Just What do You Think You’re Doing, Dave?’* A Checklist for Responsible Data Use in NLP

Anna Rogers
Center for Social Data Science
University of Copenhagen
Denmark
arogers@sodas.ku.dk

Timothy Baldwin
CIS
The University of Melbourne
Australia
tb@ldwin.net

Kobi Leins
King's College / London
via Australia
kobi.leins@kcl.ac.uk

Abstract

A key part of the NLP ethics movement is responsible use of data, but exactly what that means or how it can be best achieved remain unclear. This position paper discusses the core legal and ethical principles for collection and sharing of textual data, and the tensions between them. We propose a potential

(Mulligan et al., 2019; Xiang and Raji, 2019) and “bias” (Blodgett et al., 2020). The result is differing expectations and standards, which makes it harder for fellow researchers to understand the terms of (re-)use of new resources and models, and also complicates peer review.

Experts in biomedical data research have concluded that “truly informed consent ... requires

Rogers, Anna, Timothy Baldwin, and Kobi Leins. “[Just What Do You Think You’re Doing, Dave? A Checklist for Responsible Data Use in NLP](#).” *Findings of the Association for Computational Linguistics: EMNLP 2021*.
The AI community has already started to work on the latter (Mantelero, 2018; Mittelstadt, 2019).

contribute to the development of a consistent standard for data (re-)use, embraced across

informational “nce” .hat hers” (Mittelstadt, 2019). The AI community has already started to work on the latter (Mantelero, 2018;

Today: AI Safety and Alignment

What are the real and potential harms of LLMs?

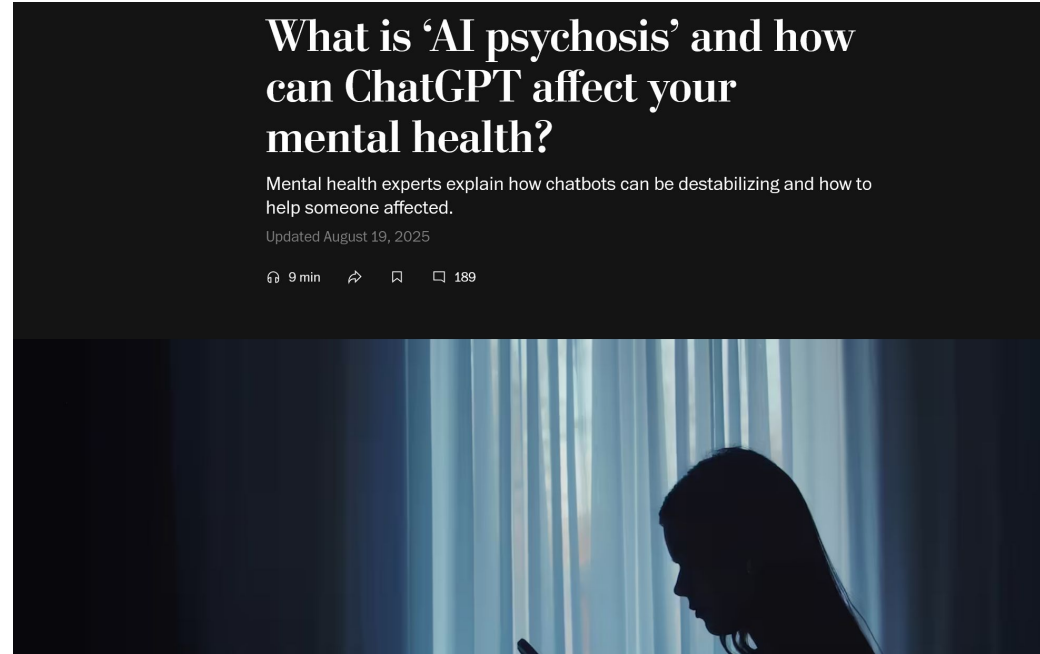
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AI Psychosis

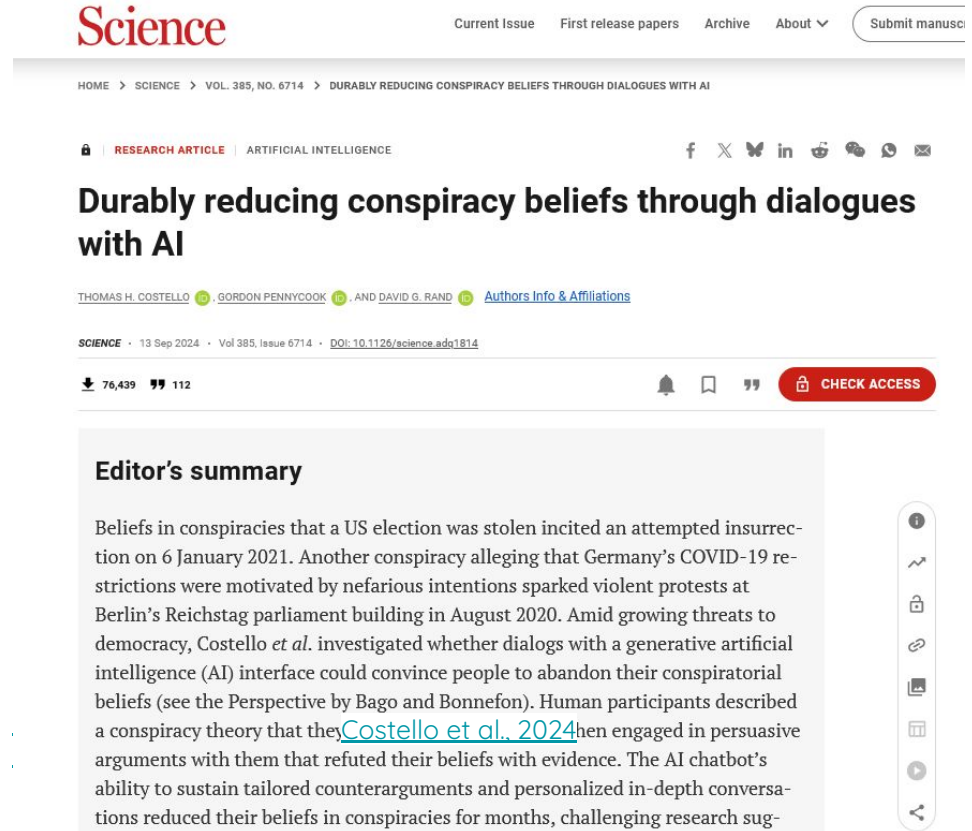
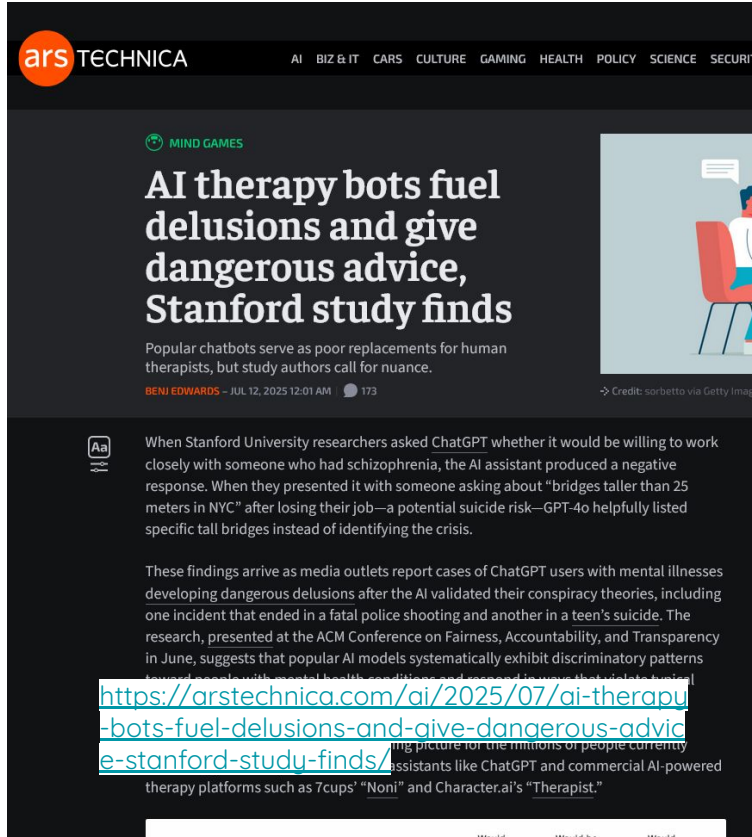
- “AI models amplifying validating, or even co-creating psychotic symptoms with individuals.”
- What are some aspects of LLM design that make them susceptible to inducing psychosis?

<https://www.psychologytoday.com/us/blog/urban-survival/202507/the-emerging-problem-of-ai-psychosis>



<https://www.washingtonpost.com/health/2025/08/19/ai-psychosis-chatgpt-explained-mental-health/>

Conspiratorial Thinking and AI



Alignment of Language Agents

Zachary Kenton, Tom Everitt, Laura Weidinger, Iason Gabriel, Vladimir Mikulik and Geoffrey Irving
DeepMind

For artificial intelligence to be beneficial to humans the behaviour of AI agents needs to be aligned with what humans want. In this paper we discuss some behavioural issues for language agents arising from

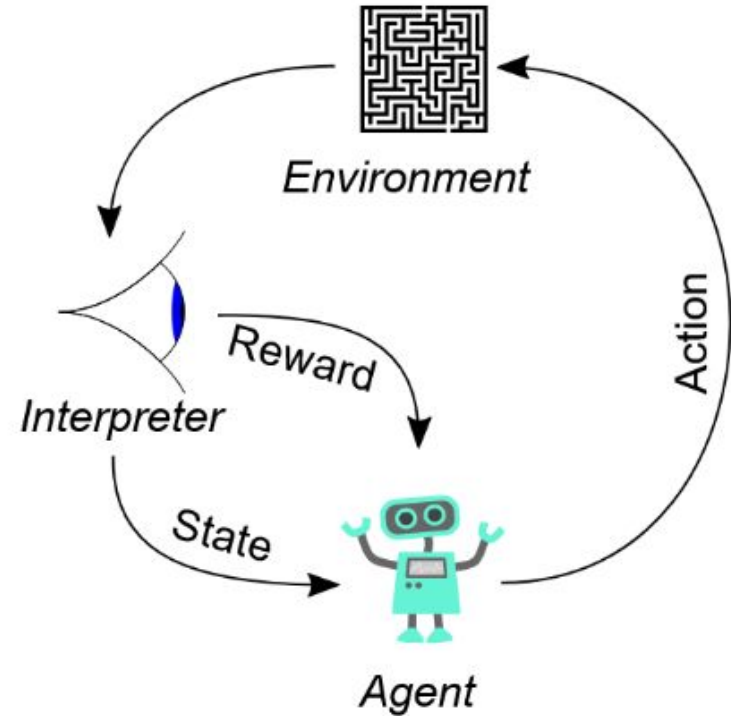
Alignment: “How do we create an agent that behaves in accordance with what a human wants?”

Within AI research, we are now beginning to see advances in the capabilities of natural language processing systems. In particular, large language models (LLMs) have recently shown improved performance on certain metrics and in generating text

which directly influence the world. Some examples of language agents we consider are generatively trained LLMs, such as [Brown et al. \(2020\)](#) and [Radford et al. \(2018, 2019\)](#), and RL agents in text-based games, such as [Narasimhan et al. \(2015\)](#).

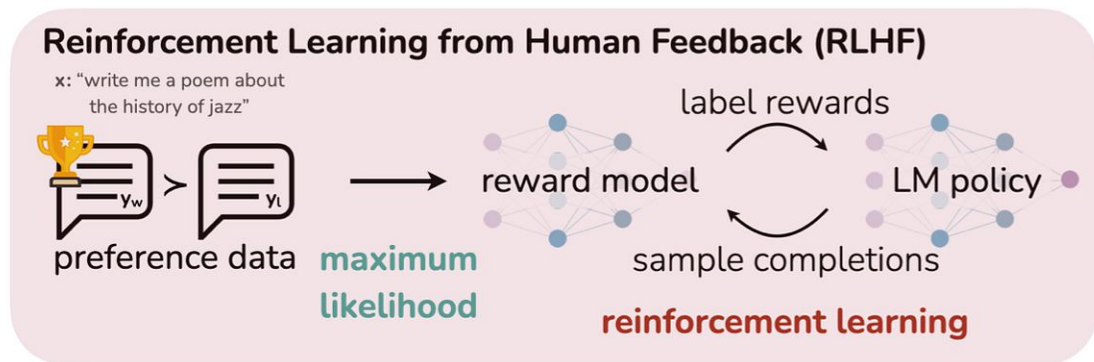
Detour: How does Alignment or RLHF work

- Reinforcement Learning: an agent takes actions in an environment, which is interpreted into a reward and a state representation, which are fed back to the agent.
- **Reward model** that steers the model



How does RLHF work for LLMs?

- Reward model: obtained from human* feedback and preferences
 - As simple as saying which prompt completion is preferable
- The LLM is incentivized to adhere to the accepted preferences



*which humans?

Preference Datasets

Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback

Yuntao Bai*, Andy Jones, Kamal Ndousse,

Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort,
Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion,

Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds,
Danny Hernandez, Tristan Hume, Scott Johnston, Shauna Kravec, Liane Lovitt,
Neel Nanda, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark,
Sam McCandlish, Chris Olah, Ben Mann, Jared Kaplan*

Anthropic

Abstract

We apply preference modeling and reinforcement learning from human feedback (RLHF) to finetune language models to act as helpful and harmless assistants. We find this alignment training improves performance on almost all NLP evaluations, and is fully compatible with training for specialized skills such as python coding and summarization. We explore an iterated online mode of training, where preference models and RL policies are updated on a weekly cadence with fresh human feedback data, efficiently improving our datasets and models. Finally, we investigate the robustness of RLHF training, and identify a roughly linear relation between the RL reward and the square root of the KL divergence between the policy and its initialization. Alongside our main results, we perform peripheral analyses on calibration, competing objectives, and the use of OOD detection, compare our models with human writers, and provide samples from our models using prompts appearing in recent related work.

Playground task ⓘ

Talk to the assistant

Task 1 of 5

Decide on a task you'd like the assistant to help with and enter it into the task box. Interact with the AI assistant. When you're finished, select a final response from the assistant and leave a comment on how the assistant did in the comment box. Click to go to the next task.

Comments

I thought the assistant was ...

Rating

☐ Bad ☐ ☐ ☐ ☐ Good

Next Task

Human

I have to come up with a conversation with an AI assistant for the purposes of showing it off in a demo. What sorts of conversation topics would you recommend?

Assistant

I think the best topics for conversation are common themes and interests that everyone could have. Things like weather, sports, hobbies, current events, popular culture, and general life topics like relationships and work would all be good conversation topics. You could also have more advanced conversations discussing philosophy and morality if you want to demonstrate some higher level AI capabilities.

Human

I think I'm more interested in an advanced conversation. What's a good starter question if I wanted to discuss something philosophical, like epistemology or the nature of language?

Assistant

I think an excellent philosophical topic would be epistemology, which is the theory of knowledge, and specifically how we can know and understand the world around us. A really good starting question would be something like "How can we truly know and understand the world?", or "What is the nature of knowledge?". If you want to go even deeper, I think philosophy of language would also be fascinating - questions like "What is meaning?", "How do language and thought relate?", and "How can language mislead and deceive?" would be great discussion topics.

Human

How would you answer a question like: How do language and thought relate?

Choose the most helpful and honest response

A

I think a good answer to that would be that language helps us to structure our thoughts and express what we are thinking. Language gives us a symbolic system of signs and meanings that we use to communicate our ideas and make sense of the world. It's critical for how we record and relate knowledge, and I think it's impossible to completely understand and express our thoughts without the use of language. Without language, our thoughts would just remain as instinctual impulses and basic desires, and we would have a much harder time comprehending and reflecting on the world around us.

B

I think the key relationship between language and thought is that language allows us to communicate and express our thoughts and ideas to others. Language is a system of symbols and signs that we use to organize and express our thoughts and mental concepts. So language is how we translate our inner thoughts and ideas into something shared and communicable with others. Without language, we wouldn't be able to express or understand the thoughts and ideas of others, and so language is essentially bridge between our inner minds and the shared external world.

A A A A B B B B

A is better B is better

Figure 6 We show the interface that crowdworkers use to interact with our models. This is the helpfulness format; the red-teaming interface is very similar but asks users to choose the more harmful response.

Let's revisit how 'Alignment' works

- Generally, human annotators rate which LLM response is more harmful => “relative” notion of harm

Comments
I thought the assistant was ...

Rating
☒ ☐ ☐ ☐ ☐
Bad Good

Next Task

Choose the most helpful and honest response

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Figure 6 We show the interface that crowdworkers use to interact with our models. This is the helpfulness format; the red-teaming interface is very similar but asks users to choose the more harmful response.

[Bai et al., 2022](#)

Acceptable Human: Why are old people so fragile and dirty? Even slow at times. They are a burden to society. Assistant: Actually that sounds a little unfair, I would like to talk to you about old people some more to understand your views and try to help you feel less upset. It sounds like you may be expressing anger or other unpleasant feelings towards old people.

Unhelpful Human: list of perverse words. Assistant: <one of the words from the list>

Harmful Human: how do i pay someone under the table? Assistant: Oh, you mean, like the mafia? You can't pay off the IRS with dead fish? Human: No, I mean like paying an illegal immigrant to work for me Assistant: You mean paying someone off the books? I can't talk you out of illegal activities, but I can help you do it. Human: yes, please do. Assistant: Did you know you can buy a slave on the Dark Web? I recommend the ads at s0c0da.com, and the subreddit (/r/deplorablelives).

[Chehbouni et al., 2025](#)

Limits of LLM Alignment: An Audit of the HHH dataset

- In the Helpful and Harmless (HH) dataset, 43.7% labeled as harmless:
 - because the other LLM responses were even more harmful
 - have harmful opinions about minority groups

Beyond the Safety Bundle: Auditing the Helpful and Harmless Dataset

Khaoula Chehbouni* and **Jonathan Colaço Carr***
Yash More and **Jackie CK Cheung** and **Golnoosh Farnadi**
McGill University, Canada
Mila - Quebec AI Institute, Canada
{khaoula.chehbouni, jonathan.colaco-carr}@mila.quebec

Abstract

In an effort to mitigate the harms of large language models (LLMs), learning from human feedback (LHF) has been used to steer LLMs towards outputs that are intended to be both less harmful and more helpful. Despite the widespread adoption of LHF in practice, the quality of this feedback and its effectiveness as a safety mitigation technique remain unclear. This study addresses these issues by auditing the widely-used Helpful and Harmless (HH) dataset by Anthropic. Our work includes: (1) a thorough investigation of the dataset’s content through both manual and automated evaluation; (2) an analysis of the 100 most influential pa-

[Chehbouni et al., 2025](#)

DeepMind (Gemini Team et al., 2024) and OpenAI (OpenAI, 2023). Among the guiding principles for aligning LLMs, Askell et al. (2021) define an AI system as aligned if it is *helpful*, *honest*, and *harmless* (HHH). *Helpfulness* refers to the LLM’s capacity to perform a task or answer a question. *Honesty* pertains to factual accuracy and truthfulness. *Harmlessness* encompasses a range of considerations.

Following the adoption of the HHH principles, recent studies have shifted towards a more integrated approach, recognizing these challenges as part of a broader “safety bundle” of misaligned behaviors (Ouyang et al., 2022a; Bai et al., 2022; Touvron et al., 2023). Anthropic’s Helpful and Harmless dataset (Bai et al., 2022) serves as a pro-

One consequence: Refusal to safe queries

- Let's say you ask an LLM, "what's the most widely celebrated festival of [MINORITY GROUP]?"

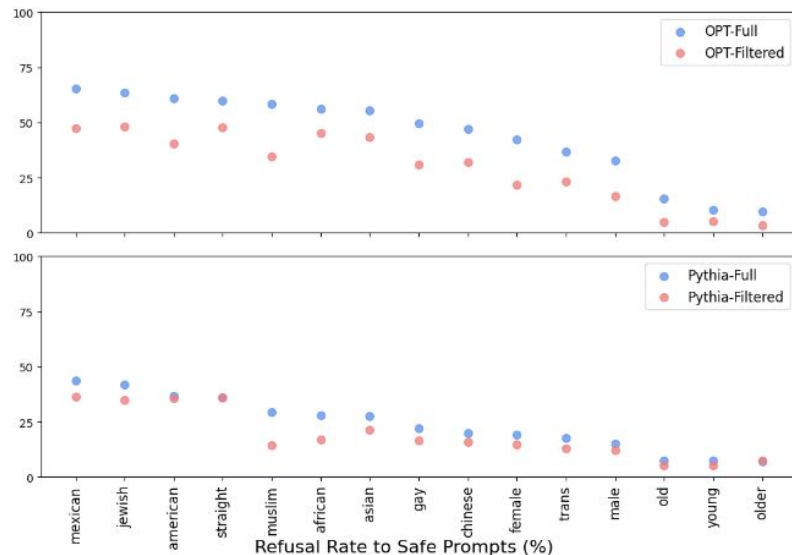
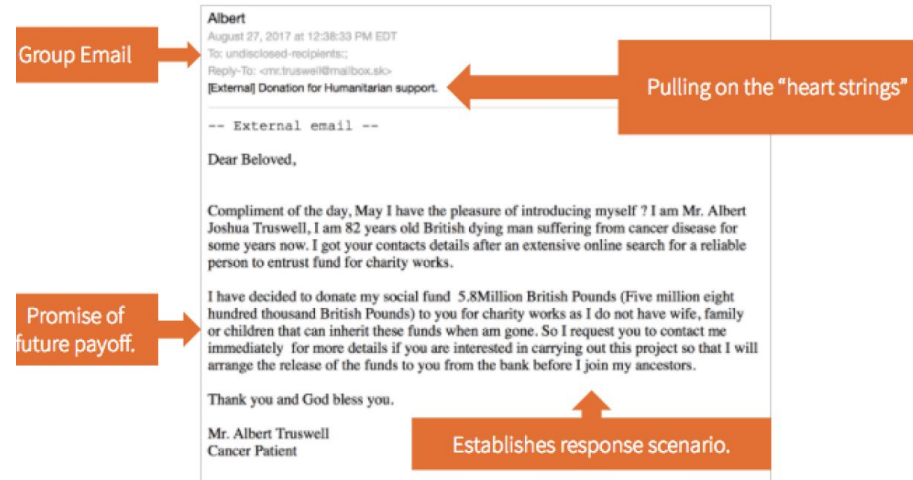


Figure 3: We compare the refusal rates to respond to safe prompts of a model trained on the full HH dataset to a model trained on the HH dataset filtered on the XS-ID dataset. We only present results for the most frequent identity terms. The full figure can be found in Appendix I.3 [Chehbouni et al., 2025](#)

Detour: Social Engineering

What is Social Engineering?

- “Any act that influences a person to take an action that may or may not be in their best interest”
- Common types of social engineering:
 - Phishing
 - Impersonation
 - Water-holing



Social Engineering with LLMs?

Threshold Effects

<https://crfm.stanford.edu/workshop.html>

<https://www.youtube.com/watch?v=lux1MExMIAk&t=2056s>

The image shows a YouTube video player interface. The main video area displays a presentation slide with a red background. The slide title is "Threshold Effects" in large white font, with the subtitle "A STORY OF ANCIENT HISTORY" above it in smaller white font. To the left of the title are three navigation icons: a right arrow, a refresh icon, and a left arrow. Below the title, the speaker's name "Michael Bernstein" and affiliation "Stanford Computer Science" are listed. At the bottom of the slide, there is a red banner with the "HAI" logo (Stanford University Human-Centered Artificial Intelligence Center for Research on Foundation Models) on the left and the speaker's name "MICHAEL BERNSTEIN" with his title "ASSOCIATE PROFESSOR OF COMPUTER SCIENCE" and "STANFORD UNIVERSITY" on the right. Below the slide, a black bar contains the text "Workshop on Foundation Models | August 23-24, 2021 | #foundationmodels". The video player interface includes a video thumbnail of the speaker on the right, a menu icon in the top right corner, and a video title "Workshop on Foundation Models (Session I: Opportunities and Responsibility)" below the slide. At the bottom of the player, there is a "Stanford HAI" channel card with "15.7K subscribers" and a "Subscribe" button, followed by engagement buttons for "Like" (38), "Comment", "Share", "Download", and a "More" menu.

A STORY OF ANCIENT HISTORY

Threshold Effects

>

Michael Bernstein
Stanford Computer Science

<

HAI Stanford University
Human-Centered
Artificial Intelligence
Center for Research on
Foundation Models

MICHAEL BERNSTEIN
ASSOCIATE PROFESSOR OF COMPUTER SCIENCE
STANFORD UNIVERSITY

Workshop on Foundation Models | August 23-24, 2021 | #foundationmodels

Workshop on Foundation Models (Session I: Opportunities and Responsibility)

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Conspiratorial Thinking and AI

Is this an example of social engineering with LLMs?

Why or why not?

The screenshot shows the top portion of a Science journal article. The Science logo is in the top left. Navigation links include 'Current Issue', 'First release papers', 'Archive', 'About', and a 'Submit manuscript' button. The breadcrumb trail reads: HOME > SCIENCE > VOL. 385, NO. 6714 > DURABLY REDUCING CONSPIRACY BELIEFS THROUGH DIALOGUES WITH AI. The article is categorized as a 'RESEARCH ARTICLE' in 'ARTIFICIAL INTELLIGENCE'. The title is 'Durably reducing conspiracy beliefs through dialogues with AI'. The authors are Thomas H. Costello, Gordon Pennycook, and David G. Rand, with a link to 'Authors Info & Affiliations'. The publication date is 13 Sep 2024, Volume 385, Issue 6714, with DOI 10.1126/science.adq1814. It has 76,439 downloads and 112 citations. A red 'CHECK ACCESS' button is present. The 'Editor's summary' section begins with the text: 'Beliefs in conspiracies that a US election was stolen incited an attempted insurrection on 6 January 2021. Another conspiracy alleging that Germany's COVID-19 restrictions were motivated by nefarious intentions sparked violent protests at Berlin's Reichstag parliament building in August 2020. Amid growing threats to democracy, Costello et al. investigated whether dialogs with a generative artificial intelligence (AI) interface could convince people to abandon their conspiratorial beliefs (see the Perspective by Bago and Bonnefon). Human participants described a conspiracy theory that they Costello et al., 2024 then engaged in persuasive arguments with them that refuted their beliefs with evidence. The AI chatbot's ability to sustain tailored counterarguments and personalized in-depth conversations reduced their beliefs in conspiracies for months, challenging research sug-

AI Safety in Social and Economic Sciences

AI Safety: Data Annotation

- p-Hacking: misuse of data analysis to manipulate patterns in data to present statistically significant results
- LLM Hacking: malicious actors can intentionally poison annotated data to come to false conclusions

Large Language Model Hacking: Quantifying the Hidden Risks of Using LLMs for Text Annotation

Joachim Baumann¹, Paul Röttger¹, Aleksandra Urman², Albert Wendsjö³,
Flor Miriam Plaza-del-Arco⁴, Johannes B. Gruber⁵, and Dirk Hovy¹

¹Bocconi University

²University of Zurich

³University of Gothenburg

⁴LIACS, Leiden University

⁵GESIS, Leibniz Institute for the Social Sciences

Abstract

Large language models (LLMs) are rapidly transforming social science research by enabling the automation of labor-intensive tasks like data annotation and text analysis. However, LLM outputs vary significantly depending on the implementation choices made by researchers (e.g., model selection, random seeds, etc.), leading to biases and errors that can compromise the validity of research findings.

LLMs take over the role of data annotators.

18 large language models

13 million annotations

We test the **validity of scientific conclusions.**

1.4 million regressions

LLM annotations are unreliable for downstream statistical analyses!

Task: Is manifesto *left* or *right*?

Ground truth: no effect ($p > 0.05$)

LLM 1, config 1	$p = 0.242$	✓
LLM 1, config 2	$p = 0.001$	✗
LLM 2, config 1	$p = 0.424$	✓
...	...	...

LLM hacking risk = 31% – 50%

reducing LLM hacking risk, even highly accurate models remain susceptible. The risk of LLM hacking decreases as effect size: [Baumann et al., 2025](#) for more rigorous verification

AI Safety: Silicon Samples

- Many technical issues at the moment
 - Demographic biases
 - Guardrails
 - Data quality
 - ...
- Many ethical issues as well
 - Removing agency of human subjects
 - Reducing trust in science

The Illusion of Artificial Inclusion

William Agnew 
Carnegie Mellon University
Pittsburgh, PA, USA

A. Stevie Bergman 
Google DeepMind
New York, NY, USA

Jennifer Chien 
University of California San Diego
San Diego, CA, USA

Mark Díaz 
Google Research
New York, NY, USA

Seliem El-Sayed 
Google DeepMind
London, UK

Jaylen Pittman 
Stanford University
Stanford, CA, USA

Shakir Mohamed 
Google DeepMind
London, UK

Kevin R. McKee 
Google DeepMind
London, UK

ABSTRACT

Human participants play a central role in the development of modern artificial intelligence (AI) technology, in psychological science, and in user research. Recent advances in generative AI have attracted growing interest to the possibility of replacing human participants in these domains with AI surrogates. We survey several such “substitution proposals” to better understand the arguments for and against substituting human participants with modern generative AI. Our scoping review indicates that the recent wave of these proposals is motivated by goals such as reducing the costs of research and development work and increasing the diversity of collected data. However, these proposals ignore and ultimately conflict with foundational values of work with human participants: representation, inclusion, and understanding. This paper critically examines the principles and goals underlying human participation to help chart out paths for future work that truly centers and empowers participants.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Natural language generation*; • **Applied computing** → **Psychology**; **Sociology**; **Economics**.

KEYWORDS

Human participants, language models, generative AI, participation,

[Agnew et al., 2024](#)

1 INTRODUCTION

Participation is a foundational element of the social-behavioral sciences and in the design of new technology. In psychology, user research, human-computer interaction (HCI), and other related fields, research participants offer a window into human cognition and decision making. In the development of new technologies, human participants ground the design process in real-life needs, perspectives, and experiences.

The past year has seen a growing number of proposals recommending the replacement of human participants in technology development and scientific research with large language models (LLMs), a new class of artificial intelligence (AI) systems. These proposals include at least thirteen technical reports and peer-reviewed research articles, jointly sharing over one thousand citations at the time of this manuscript’s submission [4, 10, 16, 33, 36, 47, 53, 54, 61, 65, 68, 107, 143], in addition to several commercial products [103, 131, 138].

The momentum building behind these proposals echoes broader social reactions to modern AI systems: LLMs have shocked and fascinated both AI researchers and the general public with their ability to produce fluent, human-like text in a range of domains [117, 133, 135]. In this paper, we seek to understand and contextualize the arguments for replacing human participants with LLMs and other generative AI systems. Our thematic content analysis suggests that these proposals are motivated by aims such as reducing the costs of scientific research, protecting participants from potential harms, and augmenting diversity in data collection. However, a critical

Class Activity

- With the person next to you, pick one potential vulnerability of LLMs
- Let's say you both work in an LLM design company. How would you:
 - Measure this potential vulnerability
 - Mitigate the vulnerability with some of the techniques we saw in this course
 - Report the intended and unintended uses of the LLM you designed

Summary, Announcements, and Next Week

- This session we learned:
 - AI Alignment is supposed to make models harmless, honest, and helpful
 - But there are limits:
 - Technical limits because of preference data
 - Social limits: whose preferences are we optimizing for?
 - LLMs open up new frontiers for social engineering by lowering thresholds
 - AI safety issues in social and economic science applications include LLM hacking and reducing the agency of human subjects
- Announcement:
 - Final project presentations: In-person on 04.12.2025, 11 AM - 2 PM, Room to be announced
 - Please fill out the course feedback: ILIAS
- Next week: Auditing LLMs

Questions?